Project 18 – Presentation Script (6 speakers, about 17 minutes)

Surface Water Quality Prediction · 14 slides · speaking pace about 130 words per minute

Plan at a glance

Speaker	Slides	Topic	Time
Mayank (Lead)	1–2	Opening and problem	1:30
Member 2	3	The CCME WQI target	1:45
Member 3	4–6	Dataset, audit, EDA	3:00
Member 4	7	Feature engineering	1:45
Mayank (Lead)	8	Pipeline and testing	1:30
Member 5	9–10	Models and benchmarks	2:45
Member 6	11–12	Validity audit and explainability	2:45
Mayank (Lead)	13–14	Limits, next steps, conclusion	1:45
Total			**about 17:00**

Fitting the 12–20 minute window: this plan lands at about 17 minutes, which leaves room for slower delivery. To cut to about 12 minutes, see the last section. To stretch to about 20, add the optional lines marked *(extra)*.

Rules for everyone: say the numbers as they appear on the slide, never say the model "predicts water quality in real time", and say "validation" every time you give a score.

Slide 1 – Title · Mayank (about 0:30)

"Good morning. We are Team 18, and our project is surface water quality prediction. We built models that estimate the Canadian CCME Water Quality Index from measured chemistry, using Random Forest, XGBoost and a small neural network. One thing we want to flag up front: besides the results, this deck includes an audit of what our very high scores really mean. We did that deliberately. Let me start with why this problem matters."

Slide 2 – The problem · Mayank (about 1:00)

"Water quality is normally checked by taking a grab sample and sending it to a lab. Some measurements are slow: BOD5 needs a five-day, 120-hour incubation. So the composite index only exists after the lab results arrive.

Our objective is a fast surrogate. Given eight measured parameters, which are pH, temperature, dissolved oxygen, BOD, ammonia, orthophosphate, nitrogen and nitrate, the model outputs the CCME index and the quality tier.

Be clear about scope. We demonstrated offline regression and classification on a historical dataset. We did not build or test real-time sensing, field deployment, or a replacement for laboratory testing. Now Member 2 will explain exactly what we are predicting."

Slide 3 – The target · Member 2 (about 1:45)

"Thank you. Our target is the CCME Water Quality Index. It is 100 minus the distance of the water from an ideal point where no guideline is violated.

The formula uses three components. F1, scope, is the percentage of variables that fail their guideline. F2, frequency, is the percentage of individual tests that fail. F3, amplitude, measures how far values exceed the guideline, through the normalised sum of excursions. We divide the combined distance by 1.732, the square root of three, so the score stays between 0 and 100.

The score maps to five tiers: Poor 0 to 44, Marginal 45 to 64, Fair 65 to 79, Good 80 to 94 and Excellent 95 to 100. In our data the mean is 85, the median 90.6, and about 72 percent of rows are Good or Excellent, so Poor is a rare class at about 4.5 percent.

Remember one point for later: the index values in the dataset are calculated from the same eight parameters that we use as inputs. Member 3 will now describe the data."

Slide 4 – Dataset · Member 3 (about 1:00)

"Our dataset has 2,827,977 raw records from 5 countries, covering 1940 to 2023. After cleaning, 2,819,617 remain, and we built 38 model features.

Look at the chart, because it shows an imbalance. England contributes about 75 percent of all rows, the USA about 14 percent, Ireland 8 percent. China is 1.6 percent and Canada has only 3,946 rows. Rivers make up 65 percent of waterbodies. So our results mainly reflect English rivers, and we cannot make country-level claims for Canada or China.

(extra) One note on sources: our report's Table I gave England as 83.7 percent, but the exploratory counts, which sum exactly to the clean total, give 75.5 percent. We use the latter and will correct the table."

Slide 5 – Data audit · Member 3 (about 1:00)

"Two different things can be wrong with data: missing values and impossible values. An older document claimed about 35 percent of values were missing. We checked all 14 columns and found zero nulls, so we did no imputation.

Impossible values are a separate matter. We applied physical bounds: pH between 3 and 12, temperature 0 to 45 degrees, dissolved oxygen 0 to 25 milligrams per litre, and no negative concentrations. This removed 8,360 rows, that is 0.296 percent, leaving 2,819,617. Most removals came from pH and temperature. The per-parameter counts add up to a little more than 8,360 because some rows break several bounds, and we removed the union."

Slide 6 – Exploratory findings · Member 3 (about 1:00)

"Two findings shaped our modelling. On the left, dissolved oxygen falls as temperature rises, following the Benson-Krause saturation curve from Henry's law. That motivated our oxygen saturation features.

On the right, nutrients such as orthophosphate are extremely skewed. Applying log of one plus x raised the Pearson correlation of orthophosphate with the index from minus 0.66 to minus 0.85, and similar gains appear for ammonia and BOD.

A caution, which is also on the slide: a better Pearson correlation is not better out-of-sample prediction. The log transform is monotonic, so rank correlation does not change, and tree models are unaffected by it. It mainly helps linear models and the neural network."

Slide 7 – Feature engineering · Member 4 (about 1:45)

"We turned 8 measurements into 38 features, grouped by the chemistry they encode.

Eight are the base sensors. Seven are aquatic stoichiometry: dissolved oxygen saturation and deficit from Benson-Krause, the Redfield nitrogen-to-phosphorus ratio that shows which nutrient limits algal growth, un-ionised ammonia from Emerson's equilibrium, which captures the toxic fraction depending on temperature and pH, and a BOD-to-oxygen ratio. Seven are log-transformed versions of skewed variables. Six are station-memory features: previous-observation lags and rolling means, sorted by area and date, and looking strictly

backwards. The lagged orthophosphate has a rank correlation of minus 0.78 with the index. Seven are temporal, including sine and cosine encodings so December sits next to January. Two are context codes: country and waterbody type.

Finally, the amber card: exceedance_counter counts guideline violations. It is built from the same type of information as the target, so we treat it as an unresolved risk, which Member 6 will cover. Over to Mayank for the pipeline."

Slide 8 – Pipeline and testing · Mayank (about 1:30)

"This is the order of our pipeline. We ingest and audit, apply physical bounds, build the 38 features, quarantine the target columns, split the data, scale on the training set only for the neural network, train and evaluate, and finally compute SHAP.

The colours matter. Navy steps happen before the split. Amber steps, the quarantine and the split, are the ones sensitive to leakage. The split is a random 70/15/15 partition, stratified by quality tier, with 1,973,731 training rows and 422,943 each for validation and test.

We also have 12 automated tests in four modules, reported as passing in 1.65 seconds. Be precise about what they show: they check bounds filtering, no look-ahead in the lags, named target columns being absent, metric maths and model loading. They do not prove that the features are free of target-derived information or that scores generalise."

Slide 9 – Modelling strategy · Member 5 (about 1:15)

"Thanks. We compared four models. A mean baseline predicts the training mean, 85.05, and is the floor any model must beat. Random Forest uses bagged, de-correlated trees. XGBoost uses boosted trees with a squared-error objective. And a dual-head neural network, 38 inputs through 256, 128 and 64 units, with one head for the index and one for the five tiers, trained for 8 epochs.

We need to be transparent about one issue. Our records of the tree settings disagree: our PDF report lists 100 trees and depth 20 for the forest and 300 rounds for XGBoost, while the repository technical report lists 30 trees and 150 estimators. The metrics are identical in both, so one description is wrong, and we will reconcile it before final submission."

Slide 10 – Benchmarking · Member 5 (about 1:30)

"All numbers here are on the validation set of 422,943 rows. The baseline has a mean absolute error of 14.13 index points. Random Forest reaches 0.052, XGBoost 0.137, and the neural network 0.706, with R-squared of 0.9999, 0.9996 and 0.9957.

So the trees are the most precise, and the neural network is about five times less precise but handles both tasks together: it classifies the five tiers with 96.1 percent accuracy and a macro F1 of 0.9485, and its probabilities are well calibrated, with an expected calibration error of 1.27 percent.

We left inference-speed numbers out on purpose, because the hardware and setup behind them are not documented. And please read the amber note: these scores measure how well each model recomputes a formula on randomly held-out rows, and Member 6 will explain what that means."

Slide 11 – Reliability and validity · Member 6 (about 1:30)

"This is the most important slide for credibility. R-squared near 0.9999 deserves careful reading, and we found three issues.

First, the index is computed from the same eight measurements that are our inputs. Our own notes describe it as an exact deterministic aggregation. So a high R-squared shows the model learned the formula.

Second, exceedance_counter counts guideline violations, the same ingredient as F1. Its low importance is expected, because the raw columns already carry that information. We could not confirm from code whether it inflates scores, so we call it an unresolved risk, not proven leakage.

Third, the split is random, and station lags are built before splitting, so neighbouring observations from one station can fall in different partitions. We have no station-held-out or time-forward test, and the test set is not yet reported.

So what can we claim? A fast, accurate surrogate of the CCME calculation on this dataset, on validation. What can we not claim? Leakage-free generalisation to new stations or future years, or independent water-quality forecasting."

Slide 12 – What drives predictions · Member 6 (about 1:15)

"On the left is built-in feature importance for Random Forest and XGBoost. Orthophosphate and its log version together account for about 88 percent of importance in the forest and 86 percent in XGBoost. The beeswarm on the right shows the SHAP view for XGBoost.

Two cautions. First, this is gain and impurity importance, not '88 percent of splits' as our report wording said. Second, importance describes the model, not the river. A plausible explanation, which we have not verified, is that the orthophosphate guideline, 0.05 milligrams per litre, sits close to the median measurement of 0.040, so it is exceeded often and moves the index most.

Also, our report said lagged orthophosphate ranks second, but it does not appear in the top features, so we removed that claim."

Slide 13 – Limitations and next steps · Mayank (about 1:00)

"Here is where we stand. Confirmed from our documents: country imbalance, a random split without station or time hold-out, validation-only metrics, and some inconsistencies between our report and repository. Open items: whether exceedance_counter inflates scores, the speed figures, possible duplicates across partitions, and that we have not yet reviewed every code file for this deck.

On status, the dual-head neural network is implemented and evaluated on validation: R-squared 0.9957 and accuracy 96.09 percent. What is planned and not done: an ablation without exceedance_counter, station-held-out and time-forward splits, reporting the test scores, and external validation."

Slide 14 – Conclusion · Mayank (about 0:45)

"To conclude. We built an audited 2.82-million-row dataset, 38 chemistry-driven features, three model families, SHAP explanations and 12 tests. Our strongest finding is that the tree models recompute the CCME index with validation error below 0.14 points. The key caveat is that the target is a formula of the inputs and the split is random, so the next step is leakage-free ablations and stricter splits. References are on the slide and the code is in our GitHub repository. Thank you, and we welcome your questions."

Cutting to about 12 minutes

- Slide 2: drop the second paragraph (0:20 saved).
- Slides 4 and 5: Member 3 merges them into one 1:15 talk, skipping the per-parameter counts (0:45 saved).
- Slide 6: skip the right-hand chart detail, keep only the caution sentence (0:30 saved).
- Slide 7: name only four groups, base, stoichiometry, station memory, and the amber card (0:45 saved).
- Slide 9: skip the neural network layer sizes and the config note, keep it for Q&A (0:30 saved).
- Slide 12: give only the first two sentences and the caution (0:30 saved).
- Slide 13: read only the *Confirmed* and *Planned* columns (0:30 saved).

Keep slides 3, 10, 11 untouched. They carry the core result and the honesty about it.

Likely questions and safe answers

Is R² = 0.9999 leakage? (Member 6) "Partly it is by construction: the index is a deterministic function of the measured inputs, so a model can learn it almost exactly. Separately, exceedance_counter is an unresolved risk we plan to test by removing it. We make no claim of independent forecasting."

Why a random split? (Member 5) "It was the protocol we used for Phase 1, stratified by tier to keep rare Poor samples in every partition. Its limitation is that station neighbours can cross partitions, so we plan station-held-out and time-forward splits."

Why did the neural network not beat the trees? (Member 5) "On this target the trees fit sharp guideline thresholds very well. The network is less precise but gives joint regression, classification and calibrated probabilities in one model."

Does log(1+x) help? (Member 4) "It raises linear correlation and helps the network. Trees are invariant to monotonic transforms, so it does not improve them."

Is orthophosphate the cause of poor water quality? (Member 6) "No. Importance is model-relative. Orthophosphate matters to the index partly because its guideline is exceeded often."

Why remove 8,360 rows instead of imputing? (Member 3) "These are impossible readings, such as negative concentrations, not missing values. There were zero nulls, so nothing needed imputing."

Could this run live on sensors? (Mayank) "We have not tested that. It is offline modelling on historical data; deployment would need validation on new stations first."

Delivery tips

- Each handover is one sentence: say "over to Member N for X" and step back, then do not repeat the previous slide.
- Practise the full run once with a timer. Aim for 16–17 minutes so you have buffer.
- Fill in the real names for Members 2–6 on slide 14 before presenting.
- Everyone should know the shared numbers: 2,827,977 raw, 2,819,617 clean, 8,360 removed, 38 features, 70/15/15 split, and the validation scores on slide 10.